

Worksheet 8 – Solution

Part 1: Time Response of First-Order Systems

1) For each of the following models, obtain the free response (zero-input response) and the time constant, if any.

a) $16\dot{x} + 14x = 0, \quad x(0) = 6$

Take Laplace transform and solve the equation for $x(t)$

$$16(sX(s) - x(0)) + 14X(s) = 0 \rightarrow X(s) = \frac{16}{16s + 14}x(0) = \frac{96}{16s + 14} = \frac{\frac{96}{14}}{\frac{16}{14}s + 1} = \frac{\frac{48}{7}}{\frac{8}{7}s + 1}$$

Free Response $\rightarrow x(t) = \frac{48}{7}e^{-\frac{7t}{8}} \rightarrow \text{Time - constant} = \frac{8}{7}$

b) $13\dot{x} - 6x = 18u_s, \quad x(0) = -2$

Take Laplace transform and solve the equation for $x(t)$

$$13(sX(s) - x(0)) - 6X(s) = 18U_s(s) \rightarrow X(s) = \frac{18}{13s - 6}U_s(s) + \frac{13}{13s - 6}x(0)$$

Free Response $\rightarrow X(s) = \frac{13}{13s - 6}x(0) \rightarrow x(t) = \frac{-26}{13s - 6} = \frac{-26/6}{\frac{13}{6}s - 1} \rightarrow x(t) = -\frac{13}{3}e^{\frac{6t}{13}}$

The system is unstable. No time-constant is defined.

c) $7\dot{x} - 5x = 0, \quad x(0) = 9$

Take Laplace transform and solve the equation for $x(t)$

$$7(sX(s) - x(0)) - 5X(s) = 0 \rightarrow X(s) = \frac{7}{7s - 5}x(0) = \frac{63}{7s - 5} = \frac{\frac{63}{5}}{\frac{7}{5}s - 1}$$

Free Response $\rightarrow x(t) = \frac{63}{5}e^{\frac{5t}{7}}$

The system is unstable. No time-constant is defined.

2) For the model $2\dot{x} + x = 10f(t)$, If $x(0) = 0$ and $f(t)$ is a unit-step, what is the steady-state response x_{ss} ? How long does it take before 98% of the difference between $x(0)$ and x_{ss} is eliminated?

Take Laplace transform and solve the equation for $X(s)$

$$2(sX(s) - x(0)) + X(s) = 10F(s) \rightarrow X(s) = \frac{10}{2s + 1}F(s) = \left(\frac{10}{2s + 1}\right)\left(\frac{1}{s}\right) = \frac{10}{s(2s + 1)}$$

The steady-state value is obtained from the final-value theorem:

$$x_{ss} = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} s \frac{10}{s(2s + 1)} = 10$$

We have to find the settling time with 2% criteria:

$$t_s = 4\tau = 4(2) = 8 \text{ sec}$$

3) Obtain the response of the model $2\dot{v} + v = f(t)$, where $f(t) = 5t$ and $v(0) = 0$. Identify the transient and steady-state responses.

Take Laplace transform having $v(0) = 0, F(s) = \frac{5}{s^2}$, and then solve the equation for $v(t)$

$$2(sV(s) - v(0)) + V(s) = F(s) \rightarrow V(s) = \frac{1}{2s+1} F(s) = \left(\frac{1}{2s+1}\right) \left(\frac{5}{s^2}\right) = \frac{5}{s^2(2s+1)}$$

Applying partial fraction expansion and inverse Laplace transform:

$$V(s) = \frac{5}{2s^2(s+0.5)} = \frac{-10}{s} + \frac{5}{s^2} + \frac{10}{s+0.5} \rightarrow v(t) = -10 + 5t + 10e^{-0.5t}, \quad t \geq 0$$

Transient response $\rightarrow v_t(t) = 10e^{-0.5t}$

Steady-state response $\rightarrow v_{ss}(t) = 5t - 10$

4) The RC circuit shown below has the parameter values $R = 3 \times 10^6 \Omega$ and $C = 1 \mu F$. If the initial capacitor voltage is 6V and the applied voltage is $v_s(t) = 12u_s(t)$, obtain the expression for the capacitor voltage response $v(t)$.

Apply KVL for the circuit:

$$v_s(t) = v_R(t) + v_c(t) \rightarrow v_s(t) = Ri(t) + v_c(t) \rightarrow v_s(t) = RC \frac{dv_c(t)}{dt} + v_c(t)$$

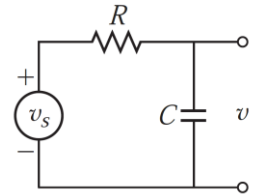
Take Laplace transform having $v_c(0) = 6$ and $V_s(s) = \frac{12}{s}$:

$$V_s(s) = RC(sV_c(s) - v_c(0)) + V_c(s) \rightarrow V_c(s) = \frac{1}{RCs+1} V_s(s) + \frac{RC}{RCs+1} v_c(0)$$

$$V_c(s) = \left(\frac{1}{3s+1}\right) \left(\frac{12}{s}\right) + \left(\frac{3}{3s+1}\right) (6) = \frac{12}{s(3s+1)} + \frac{18}{3s+1}$$

Take inverse Laplace transform:

$$v_c(t) = 12 \left(1 - e^{-\frac{t}{3}}\right) + 18e^{-\frac{t}{3}}, \quad t \geq 0$$



5) Consider the model: $6\dot{v} + 3v = \dot{g}(t) + g(t)$

where $v(0) = 0$. Obtain the response $v(t)$ if $g(t) = u_s(t)$.

Take Laplace transform having $v(0) = 0, g(0) = 0$ and $G(s) = \frac{1}{s}$:

$$6(sV(s) - v(0)) + 3V(s) = sG(s) - g(0) + G(s) \rightarrow 6sV(s) + 3V(s) = s \left(\frac{1}{s}\right) + \frac{1}{s}$$

$$V(s) = \frac{s+1}{s(6s+3)} = \frac{s+1}{6s(s+0.5)} = \frac{1/3}{s} + \frac{-1/6}{s+0.5}$$

Take inverse Laplace transform:

$$v(t) = \frac{1}{3} - \frac{1}{6} e^{-0.5t}, \quad t \geq 0$$

6) Obtain the response of the model $9\dot{v} + 3v = f(t)$, where $f(t) = 7t$ and $v(0) = 0$. Is the steady-state response parallel to $f(t)$?

Take Laplace transform having $v(0) = 0, g(0) = 0$ and $F(s) = \frac{7}{s^2}$:

$$9(sV(s) - v(0)) + 3V(s) = F(s) \rightarrow 9sV(s) + 3V(s) = \frac{7}{s^2}$$

$$V(s) = \frac{7}{s^2(9s + 3)} = \frac{7}{9s^2(s + 1/3)} = \frac{7}{s} + \frac{7/3}{s^2} + \frac{7}{s + 1/3}$$

Take inverse Laplace transform:

$$v(t) = 7 + \frac{7}{3}t + 7e^{-t/3}, \quad t \geq 0$$

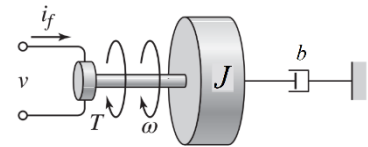
Steady-state response $\rightarrow v_{ss}(t) = 7 + \frac{7}{3}t$ which is not parallel to the input $7t$ because the slope is not same.

7) A certain rotational system has inertia $J = 20 \text{ kg.m}^2$ and a viscous damping constant $b = 10 \text{ N.m.s/rad}$. The torque $T(t)$ is applied by an electric motor. The equation of motion is:

$$50 \frac{d\omega}{dt} + 10\omega = \tau(t)$$

The model of the motor's field current i_f in amperes is:

$$0.001 \frac{di_f}{dt} + 5i_f = v(t)$$



where $v(t)$ is the voltage applied to the motor. The motor torque constant is $K_T = 25 \text{ N.m/A}$. Suppose the applied voltage is 10V. Determine the steady-state speed of the inertia and estimate the time required to reach that speed.

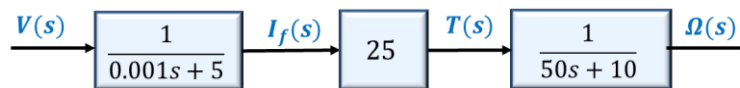
Transfer function of the mechanical subsystem is:

$$50 \frac{d\omega}{dt} + 10\omega = \tau(t) \rightarrow 50s\Omega(s) + 10\Omega(s) = T(s) \rightarrow \frac{\Omega(s)}{T(s)} = \frac{1}{50s + 10}$$

Transfer function of the electrical subsystem is:

$$0.001 \frac{di_f}{dt} + 5i_f = v(t) \rightarrow 0.001sI_f(s) + 5I_f(s) = V(s) \rightarrow \frac{I_f(s)}{V(s)} = \frac{1}{0.001s + 5}$$

The overall block diagram model and the overall transfer function model of the system is,



$$\frac{\Omega(s)}{V(s)} = \frac{25}{(0.001s + 5)(50s + 10)}$$

The steady-state speed is obtained using the final-value theorem:

$$\lim_{t \rightarrow \infty} \omega(t) = \lim_{s \rightarrow 0} s\Omega(s) \rightarrow \omega_{ss} = \omega(\infty) = \lim_{s \rightarrow 0} s \left(\frac{25}{(0.001s + 5)(50s + 10)} \right) \left(\frac{10}{s} \right) = 5 \text{ rad/s}$$

Time constant of the electrical subsystem is 0.001 sec.

Time constant of the mechanical subsystem is 5 sec.

Since the electrical subsystem is much **faster** than the mechanical subsystem, the overall time constant can be estimated as the time constant of the mechanical subsystem.

Therefore, the estimated time to reach the final speed is about $4\tau = 4(5) = 20 \text{ sec}$

Part 2: Time Response of Second-Order Systems

1) If applicable, compute ζ, τ, ω_n and ω_d for the following roots, and find the corresponding characteristic polynomial.

Having the general form of the second-order system characteristic equation and the pole locations:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad s = -\sigma \pm j\omega_d = -\zeta\omega_n \pm \omega_n\sqrt{1-\zeta^2}$$

a) $s = -2 \pm j6$

The characteristic polynomial is: $(s + 2 + j6)(s + 2 - j6) = s^2 + 4s + 40$

The system is stable with complex poles.

$$\omega_n = \sqrt{40}, \quad \zeta = \frac{2}{\sqrt{40}} = 0.316, \quad \omega_d = 6$$

b) $s = -10, -10,$

The characteristic polynomial is: $(s + 10)(s + 10) = s^2 + 20s + 100$

The system is stable with repetitive real poles.

$$\zeta = 1$$

ω_n and ω_d are not defined because the free response is not oscillatory.

c) $s = -10,$

The characteristic polynomial is: $s + 10$

The system is first order, so ζ, ω_n and ω_d are not defined because the free response is not oscillatory.

Time constant is $\tau = 1/10$

d) $s = 1 \pm j5$

The characteristic polynomial is: $(s - 1 + j5)(s - 1 - j5) = s^2 - 2s + 26$

The system is unstable, so damping ratio ζ and time-constant τ are not defined.

$$\omega_d = 5, \quad \omega_n = \sqrt{26}$$

2) Find the response for the following models. The initial conditions are zero.

a) $3\ddot{x} + 21\dot{x} + 30x = 4t$

Take Laplace transform having all initial conditions are zero:

$$3s^2X(s) + 21sX(s) + 30X(s) = \frac{4}{s^2} \quad \rightarrow \quad X(s) = \frac{4}{3s^2(s^2 + 7s + 10)} = \frac{4}{3s^2(s+2)(s+5)}$$

$$X(s) = \frac{4}{3} \left(\frac{0.1}{s^2} - \frac{0.07}{s} + \frac{0.0833}{s+2} - \frac{0.0133}{s+5} \right)$$

Take inverse Laplace transform:

$$x(t) = \frac{4}{3} (0.1t - 0.07 + 0.0833e^{-2t} - 0.0133e^{-5t}), \quad t \geq 0$$

b) $5\ddot{x} + 20\dot{x} + 20x = 7t$

Take Laplace transform having all initial conditions are zero:

$$5s^2X(s) + 20sX(s) + 20X(s) = \frac{7}{s^2} \quad \rightarrow \quad X(s) = \frac{7}{5s^2(s^2 + 4s + 4)} = \frac{7}{5s^2(s+2)^2}$$

$$X(s) = \frac{7}{5} \left(\frac{0.25}{s^2} - \frac{0.25}{2} + \frac{0.25}{(s+2)^2} + \frac{0.25}{s+2} \right)$$

Take inverse Laplace transform:

$$x(t) = \frac{7}{5} (0.25t - 0.25 + 0.25te^{-2t} + 0.25e^{-2t}), \quad t \geq 0$$

c) $2\ddot{x} + 8\dot{x} + 58x = 5t$

Take Laplace transform having all initial conditions are zero:

$$2s^2X(s) + 8sX(s) + 58X(s) = \frac{5}{s^2} \quad \rightarrow \quad X(s) = \frac{5}{2s^2(s^2 + 4s + 29)} = \frac{5}{2s^2((s+2)^2 + 25)}$$

$$X(s) = \frac{5}{2} \left(\frac{\frac{1}{29}}{s^2} - \frac{\frac{4}{29^2}}{s} - \frac{\frac{21}{29^2}}{(s+2)^2 + 25} + \frac{\frac{4}{29^2}(s+2)}{(s+2)^2 + 25} \right)$$

Take inverse Laplace transform:

$$x(t) = \frac{5}{2(29)^2} \left(29t - 4 + \frac{21}{5} e^{-2t} \sin 5t + 4e^{-2t} \cos 5t \right), \quad t \geq 0$$

3) The characteristic equation of a certain system is $4s^2 + 6ds + 25d^2 = 0$, where d is a constant.

a) For what values of d is the system stable?

Having the general form of the second-order system characteristic equation and the pole locations:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad s = -\sigma \pm j\omega_d = -\zeta\omega_n \pm \omega_n\sqrt{1-\zeta^2}$$

The system is stable if $\zeta\omega_n > 0$ and $\omega_n^2 > 0$.

The characteristic equation in standard form is: $s^2 + \frac{6}{4}ds + \frac{25}{4}d^2 = 0$

So, to have a stable system we must have $\frac{3}{4}d > 0$ and $\frac{25}{4}d^2 > 0$ or simply $d > 0$.

b) Is there a value of d for which the free response will consist of decaying oscillations?

The damping ratio is obtained as:

$$2\zeta\omega_n = \frac{6}{4}d \quad \rightarrow \quad \zeta = \frac{\frac{6}{4}d}{2\omega_n} = \frac{\frac{6}{4}d}{2\left(\frac{5}{2}d\right)} = \frac{6}{20} < 1$$

Thus there will be damped oscillations for any $d > 0$.

4) The characteristic equation of a certain system is $s^2 + 6bs + 5b - 10 = 0$, where b is a constant.

a) For what values of b is the system stable?

Having the general form of the second-order system characteristic equation and the pole locations:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad s = -\sigma \pm j\omega_d = -\zeta\omega_n \pm \omega_n\sqrt{1 - \zeta^2}$$

The system is stable if $\zeta\omega_n > 0$ and $\omega_n^2 > 0$.

The characteristic equation in standard form is: $s^2 + 6bs + (5b - 10) = 0$

So, to have a stable system we must have $3b > 0$ and $5b - 10 > 0$ or simply $b > 2$.

b) Is there a value of b for which the free response will consist of decaying oscillations?

The damping ratio is obtained as:

$$2\zeta\omega_n = 6b \quad \rightarrow \quad \zeta = \frac{6b}{2(5b - 10)} < 1 \quad \text{for all } b > 5$$

Thus there will be damped oscillations for $b > 5$.

5) Given the model: $\ddot{x} - (\mu + 2)\dot{x} + (2\mu + 5)x = 0$

a) Find the values of the parameter μ for which the system is: Stable, Marginally Stable and Unstable.

Having the general form of the second-order system characteristic equation and the pole locations:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad s = -\sigma \pm j\omega_d = -\zeta\omega_n \pm \omega_n\sqrt{1 - \zeta^2}$$

The system is stable if $\zeta\omega_n > 0$ and $\omega_n^2 > 0$.

The characteristic equation in standard form is: $s^2 - (\mu + 2)s + (2\mu + 5) = 0$

To have a stable system we must have $\frac{-(\mu+2)}{2} > 0$ and $2\mu + 5 > 0$ or simply $-2.5 < \mu < -2$.

To have a marginally stable system we must have $-(\mu + 2) = 0$ or simply $\mu = -2$.

To have an unstable system we must have $\mu > -2$ and $\mu < -2.5$.

6) A certain armature-controlled DC motor has the characteristic equation

$$L_a J s^2 + (R_a J + b L_a) s + b R_a + K_b K_T = 0$$

Using the following parameter values:

$$K_b = K_T = 0.1 \text{ N.m/A}, \quad J = 6 \times 10^{-5} \text{ kg.m}^2, \quad R_a = 0.6 \Omega, \quad L_a = 4 \times 10^{-3} \text{ H}$$

a) Obtain the expression for the damping ratio ζ and undamped natural frequency ω_n in terms of the damping b .

The characteristic equation becomes:

$$(24 \times 10^{-8}) s^2 + (3.6 \times 10^{-5} + 4b \times 10^{-3}) s + 0.6b + 0.01 = 0$$

Comparing with the standard form of $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$,

$$s^2 + \frac{(3.6 \times 10^{-5} + 4b \times 10^{-3})}{24 \times 10^{-8}} s + \frac{0.6b + 0.01}{24 \times 10^{-8}} = 0$$

The undamped natural frequency is:

$$\omega_n = \sqrt{\frac{0.6b + 0.01}{24 \times 10^{-8}}} = 5000 \sqrt{\frac{0.6b + 0.01}{6}}$$

The damping ratio is:

$$2\zeta\omega_n = \frac{(3.6 \times 10^{-5} + 4b \times 10^{-3})}{24 \times 10^{-8}} \rightarrow \zeta = \frac{3.6 \times 10^{-5} + 4b \times 10^{-3}}{2(24 \times 10^{-8})5000 \sqrt{\frac{0.6b + 0.01}{6}}} = \frac{0.09 + 10b}{\sqrt{6(0.6b + 0.01)}}$$

b) Find the required damping b to have a critically-damped response.

$$\zeta = 1 \rightarrow 1 = \frac{0.09 + 10b}{\sqrt{6(0.6b + 0.01)}} \rightarrow \sqrt{6(0.6b + 0.01)} = 0.09 + 10b$$

$$6(0.6b + 0.01) = 0.0081 + 100b^2 + 1.8b \rightarrow 100b^2 - 1.8b - 0.0519 = 0$$

Solving the quadratic equation we have: $b = 0.0335$ and $b = -0.0155$

Since the damping value should be a positive number, the required damping is: $b = 3.35 \times 10^{-2} \text{ N.s/m}$

Part 3: Description and Specification of Step Response

1) Compute the maximum percent overshoot, maximum overshoot, peak time, rise time and settling time (2%) for the following model. The initial conditions are zero. Time is measured in second.

$$\ddot{y} + 4\dot{y} + 8y = 2u_s(t)$$

The system transfer function is obtained as:

$$s^2 Y(s) + 4s Y(s) + 8Y(s) = 2U(s) \rightarrow \frac{Y(s)}{U(s)} = \frac{2}{s^2 + 4s + 8}$$

The characteristic equation is: $s^2 + 4s + 8 = 0$

The standard form of the second-order system characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The undamped natural frequency is:

$$\omega_n^2 = 8 \rightarrow \omega_n = \sqrt{8}$$

The damping ratio is:

$$2\zeta\omega_n = 4 \rightarrow \zeta = \frac{4}{2\sqrt{8}} = \frac{1}{\sqrt{2}} = 0.707$$

The maximum overshoot and its percent are:

$$O.S. \% = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\% = e^{-0.707\pi/\sqrt{1-0.707^2}} \times 100\% = 0.043 \times 100\% \rightarrow O.S. \% = 4.3\%$$

The peak-time is:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{(\sqrt{8})\sqrt{1-0.707^2}} = 1.57 \text{ sec}$$

The rise-time is:

$$t_r \approx \frac{0.8 + 2.5\zeta}{\omega_n} = \frac{0.8 + 2.5(0.707)}{\sqrt{8}} = 0.91 \text{ sec}$$

The 2% settling-time is:

$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{0.707\sqrt{8}} = 2.00 \text{ sec}$$

2) A certain system is described by the model

$$\ddot{x} + b\dot{x} + 4x = u_s(t)$$

Set the value of the damping constant b so that both of the following specifications are satisfied. Give priority to the overshoot specifications. If both cannot be satisfied, state the reason. Time is measured in seconds.

a) Maximum percent overshoot should be as small as possible and no greater than 20%.

b) Rise time should be as small as possible and no greater than 3s.

The system transfer function is obtained as:

$$s^2X(s) + bsX(s) + 4X(s) = U(s) \rightarrow \frac{X(s)}{U(s)} = \frac{1}{s^2 + bs + 4}$$

The characteristic equation is: $s^2 + bs + 4 = 0$

The standard form of the second-order system characteristic equation is:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The undamped natural frequency is:

$$\omega_n^2 = 4 \rightarrow \omega_n = 2$$

The damping ratio is:

$$2\zeta\omega_n = b \rightarrow \zeta = \frac{b}{4}$$

To have a maximum overshoot less than 20%, the damping ratio must be greater than 0.46:

$$\zeta = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}} = \frac{-\ln(0.2)}{\sqrt{\pi^2 + \ln^2(0.2)}} \rightarrow \zeta = 0.46$$

To have a rise-time less than 3sec, the damping ratio must be less than 2.08:

$$t_r \cong \frac{0.8 + 2.5\zeta}{\omega_n} = 3 \rightarrow \zeta = \frac{3\omega_n - 0.8}{2.5} = \frac{3(2) - 0.8}{2.5} = 2.08$$

Therefore, the damping ratio must be $0.46 \leq \zeta \leq 2.08$.

The required damping constant range is obtained as:

$$0.46 \leq \frac{b}{4} \leq 2.08 \rightarrow 1.84 \leq b \leq 8.32$$

3) A certain system is described by the model

$$9\ddot{x} + b\dot{x} + 4x = u_s(t)$$

Set the value of the damping constant b so that both of the following specifications are satisfied. Give priority to the overshoot specifications. If both cannot be satisfied, state the reason. Time is measured in seconds.

a) Maximum percent overshoot should be as small as possible and no greater than 20%.

b) Rise time should be as small as possible and no greater than 3s.

The system transfer function is obtained as:

$$9s^2X(s) + bsX(s) + 4X(s) = U(s) \rightarrow \frac{X(s)}{U(s)} = \frac{1}{9s^2 + bs + 4} = \frac{\frac{1}{9}}{s^2 + \frac{b}{9}s + \frac{4}{9}}$$

The characteristic equation is: $s^2 + \frac{b}{9}s + \frac{4}{9} = 0$

The standard form of the second-order system characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The undamped natural frequency is:

$$\omega_n^2 = \frac{4}{9} \rightarrow \omega_n = \frac{2}{3}$$

The damping ratio is:

$$2\zeta\omega_n = \frac{b}{9} \rightarrow \zeta = \frac{b}{18\omega_n} = \frac{b}{18\left(\frac{2}{3}\right)} = \frac{b}{12}$$

To have a maximum overshoot less than 20%, the damping ratio must be greater than 0.46:

$$\zeta = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}} = \frac{-\ln(0.2)}{\sqrt{\pi^2 + \ln^2(0.2)}} \rightarrow \zeta = 0.46$$

To have a rise-time less than 3sec, the damping ratio must be less than 0.48:

$$t_r \cong \frac{0.8 + 2.5\zeta}{\omega_n} = 3 \rightarrow \zeta = \frac{3\omega_n - 0.8}{2.5} = \frac{3\left(\frac{2}{3}\right) - 0.8}{2.5} = 0.48$$

Therefore, the damping ratio must be $0.46 \leq \zeta \leq 0.48$.

The required damping constant range is obtained as:

$$0.46 \leq \frac{b}{12} \leq 0.48 \rightarrow 5.52 \leq b \leq 5.76$$

4) Derive the fact that the peak time is the same for all characteristic roots having the same imaginary part.

The peak time is calculated by the following formula:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

The general form of the characteristic roots for a second-order under-damped system is:

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{\zeta^2 - 1}$$

Comparing the peak time formula and the roots we can see that the peak time depends only on the imaginary part of the characteristic roots.

5) Derive the fact that the settling time is the same for all characteristic roots having the same real part.

The settling time (2%) is calculated by the following formula:

$$t_s = \frac{4}{\zeta\omega_n}$$

The general form of the characteristic roots for a second-order under-damped system is:

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{\zeta^2 - 1}$$

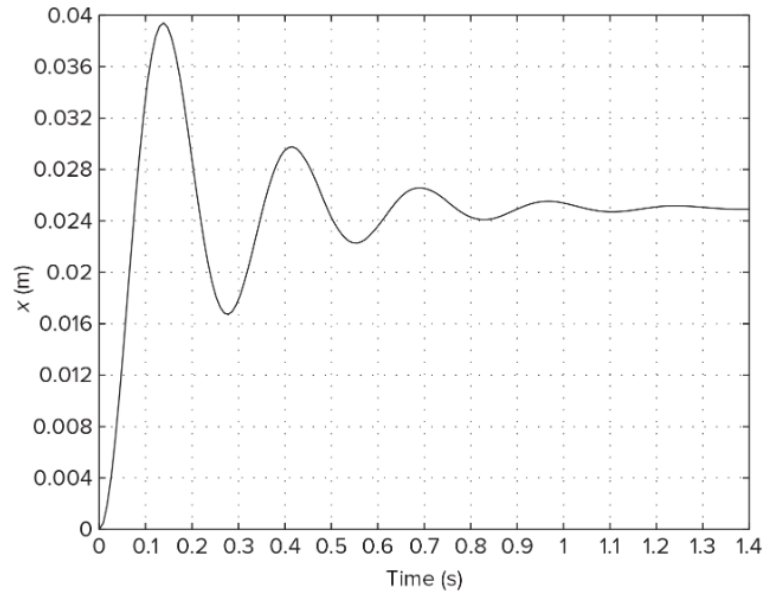
Comparing the settling time formula and the roots we can see that the settling time depends only on the real part of the characteristic roots.

Part 4: System Modeling via Transient Response

- 1) The following figure shows the response of a system to a step input of magnitude 1000 N. The equation of motion is:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$

Estimate the values of m , b , and k .



From the equation of motion, the transfer function model of the system is obtained as:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \quad \rightarrow \quad G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

Since the model is a second-order transfer function, we can find the model parameters in terms of the **DC-gain**, **damping ratio** and **undamped natural frequency** and determine those values from the step response and estimate the system parameters.

$$G(s) = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}} \quad \text{and} \quad G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

Comparing the transfer function with the standard form we have:

$$k = \frac{1}{K}, \quad m = \frac{1}{K\omega_n^2}, \quad b = \frac{2\zeta}{K\omega_n}$$

From the given step response we can find the steady-state value, peak-time and maximum deviation from the steady-state value:

$$x_{ss} = 0.025 \text{ m}, \quad t_p = 0.14 \text{ sec}, \quad M_p = 0.038 - 0.025 = 0.013 \text{ m}$$

The DC-gain of system is obtained as:

$$K = \frac{\Delta x}{\Delta f} = \frac{x_{ss} - x_0}{f_{ss} - f_0} = \frac{0.025 - 0}{1000 - 0} = 2.5 \times 10^{-5} \text{ m/N}$$

The damping ratio is determined as:

$$O.S. = \frac{M_p}{x_{ss}} = \frac{0.013}{0.025} = 0.52$$

$$\zeta = \frac{-\ln(O.S.)}{\sqrt{\pi^2 + \ln^2(O.S.)}} = \frac{-\ln(0.52)}{\sqrt{\pi^2 + \ln^2(0.52)}} = 0.204$$

The undamped natural frequency is calculated as:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \rightarrow 0.14 = \frac{\pi}{\omega_n \sqrt{1 - (0.204)^2}} \rightarrow \omega_n = 22.92 \text{ rad/s}$$

The system parameters are estimated as:

$$k = \frac{1}{K} \rightarrow k = \frac{1}{2.5 \times 10^{-5}} = 40000 \text{ N/m}$$

$$m = \frac{1}{K \omega_n^2} \rightarrow m = \frac{1}{2.5 \times 10^{-5} (22.92)^2} = 76.14 \text{ kg}$$

$$b = \frac{2\zeta}{K \omega_n} \rightarrow b = \frac{2(0.204)}{2.5 \times 10^{-5} (22.92)} = 698.1 \text{ N.s/m}$$

- 2) For the following model, find the steady-state response and use the dominant pole approximation to find the dominant response (how long will take to reach steady-state? does it oscillate?). The initial conditions are zero. ($u_s(t)$ is the unit-step signal).

$$\ddot{x}(t) + 22\dot{x}(t) + 131x(t) = u_s(t)$$

Obtain the exact solution for the response of the above model, and use it to check the prediction based on the dominant pole approximation.

We can find the transfer function and the characteristics equation from the given differential equation:

$$s^3 X(s) + 22s^2 X(s) + 131s X(s) + 110X(s) = U_s(s)$$

$$G(s) = \frac{X(s)}{U_s(s)} = \frac{1}{s^3 + 22s^2 + 131s + 110}$$

Since, $u_s(t)$ is a unit-step input the steady-state response is:

$$x_{ss} = \lim_{s \rightarrow 0} sG(s) = s \left(\frac{1}{s^3 + 22s^2 + 131s + 110} \right) \left(\frac{1}{s} \right) = \frac{1}{110} = 0.0091$$

The characteristics equation and the poles are:

$$s^3 + 22s^2 + 131s + 110 = 0 \rightarrow s = -1, \quad s = -10, \quad s = -11$$

The dominant pole is $s = -1$ and the dominant time constant is 1sec.

Since all poles are real, there will be no oscillations in the response, and the steady-state response will be reached at approximately $t_s \approx 4(1 \text{ sec}) = 4 \text{ sec}$

We can apply the partial fraction expansion to find the exact solution:

$$X(s) = \frac{1}{s^3 + 22s^2 + 131s + 110} U_s(s) = \frac{1}{(s+1)(s+10)(s+11)s} = \frac{A}{s+1} + \frac{B}{s+10} + \frac{C}{s+11} + \frac{D}{s}$$

$$A = \left[\frac{1}{(s+10)(s+11)s} \right]_{s=-1} = -0.011$$

$$B = \left[\frac{1}{(s+1)(s+11)s} \right]_{s=-10} = 0.011$$

$$C = \left[\frac{1}{(s+1)(s+10)s} \right]_{s=-11} = -0.0091$$

$$D = \left[\frac{1}{(s+1)(s+10)(s+11)} \right]_{s=0} = 0.0091$$

$$X(s) = -\frac{0.011}{s+1} + \frac{0.011}{s+10} - \frac{0.0091}{s+11} + \frac{0.0091}{s}$$

Taking the inverse Laplace transform the exact solution is:

$$x(t) = -0.011e^{-t} + 0.011e^{-10t} - 0.0091e^{-11t} + 0.0091$$

From the exact solution we can also find the steady-state response as:

$$x_{ss} = \lim_{t \rightarrow \infty} x(t) = 0.0091$$

- 3) The following model has a dominant poles of $s = -3 \pm j5$ as long as the parameter μ is no less than 3. ($u_s(t)$ is the unit-step signal).

$$\ddot{y}(t) + (6 + \mu)\dot{y}(t) + (34 + 6\mu)y(t) = u_s(t)$$

Investigate the accuracy of the estimate of the maximum overshoot, the peak time, the rise time, and the 2% settling time based on the dominant root, for the following cases: (a) $\mu = 30$, (b) $\mu = 6$, and $\mu = 3$. Discuss the effect of μ on these predictions.

The factor of the given dominant poles pair is:

$$(s + 3 + j5)(s + 3 - j5) = s^2 + 6s + 34$$

To find the third pole $s = -p$, we can expand the equation as follows:

$$(s^2 + 6s + 34)(s + p) = s^3 + (6 + p)s^2 + (34 + 6p)s + 34p$$

Next, we can find the characteristic polynomial from the given differential equation and compare it with the obtained one:

$$s^3 Y(s) + (6 + \mu)s^2 Y(s) + (34 + 6\mu)s Y(s) + 34\mu Y(s) = U_s(s)$$

The characteristic polynomial is:

$$s^3 + (6 + \mu)s^2 + (34 + 6\mu)s + 34 = 0$$

We can see that the third pole is $s = -\mu$. Thus, as long as $\mu > 3$, the poles at $s = -3 \pm j5$ will be closer to the imaginary axis, which means is dominant.

We can find the damping ratio and the undamped natural frequency from the dominant poles:

$$s = -3 \pm j5 = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$$

The damping ratio is obtained as:

$$\zeta = \cos \theta = \cos \left[\tan^{-1} \left(\frac{\text{Im}[s]}{\text{Re}[s]} \right) \right] = \cos \left[\tan^{-1} \left(\frac{5}{3} \right) \right] = 0.514$$

The undamped natural frequency is obtained as:

$$\omega_n = \sqrt{\text{Re}[s]^2 + \text{Im}[s]^2} = \sqrt{9 + 25} = \sqrt{34} = 5.83 \text{ rad/s}$$

The settling time from the dominant poles is about:

$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{3} = 1.33 \text{ sec}$$

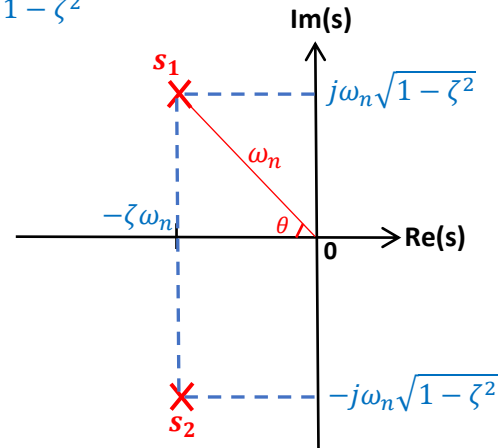
The maximum overshoot is estimated as:

$$O.S. \% = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = 0.15 \times 100\% = 15\%$$

The peak-time and rise-time are estimated as:

$$t_p = \frac{\pi}{\omega_n\sqrt{1 - \zeta^2}} = \frac{\pi}{5.83\sqrt{1 - (0.514)^2}} = 0.63 \text{ sec}$$

$$t_r = \frac{0.8 + 2.5\zeta}{\omega_n} = \frac{0.8 + 2.5(0.514)}{5.83} = 0.36 \text{ sec}$$



The above calculations are based on the dominant poles approximation, which is less accurate when the root separation factor is small. Thus the calculations for $\mu = 30$ have the most accuracy.

- 4) Estimate the maximum overshoot, the peak time, and the rise time of the unit-step response of the following model if $f(t) = 5000u_s(t)$ and the initial conditions are zero.

$$\frac{d^4 y}{dt^4} + 26 \frac{d^3 y}{dt^3} + 269 \frac{d^2 y}{dt^2} + 1524 \frac{dy}{dt} + 4680 y(t) = f(t)$$

Its poles are $s = -3 \pm j6$ and $s = -10 \pm j2$.

The dominant pole pair is $s = -3 \pm j6$.

We can find the damping ratio and the undamped natural frequency from the dominant poles:

$$s = -3 \pm j6 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The damping ratio is obtained as:

$$\zeta = \cos \theta = \cos \left[\tan^{-1} \left(\frac{\text{Im}[s]}{\text{Re}[s]} \right) \right] = \cos \left[\tan^{-1} \left(\frac{6}{3} \right) \right] = 0.447$$

The undamped natural frequency is obtained as:

$$\omega_n = \sqrt{\text{Re}[s]^2 + \text{Im}[s]^2} = \sqrt{9 + 36} = \sqrt{45} = 6.708 \text{ rad/s}$$

The settling time from the dominant poles is about:

$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{3} = 1.33 \text{ sec}$$

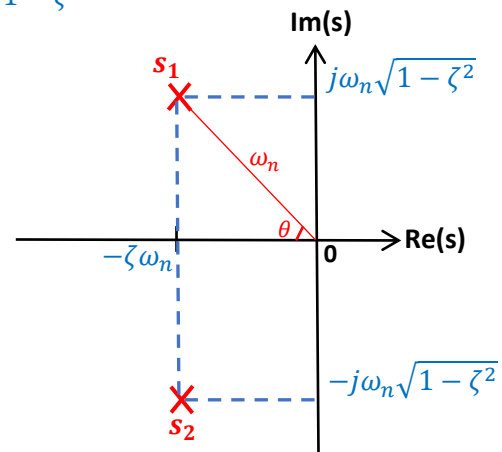
The maximum overshoot is estimated as:

$$O.S. \% = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = 0.21 \times 100\% = 21\%$$

The peak-time and rise-time are estimated as:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{6.708\sqrt{1-(0.447)^2}} = 0.524 \text{ sec}$$

$$t_r = \frac{0.8 + 2.5\zeta}{\omega_n} = \frac{0.8 + 2.5(0.447)}{6.708} = 0.286 \text{ sec}$$



- 5) What is the form of the unit-step response of the following model? Find the steady-state response. How long does the response take to reach steady-state?

$$2 \frac{d^4 y}{dt^4} + 52 \frac{d^3 y}{dt^3} + 6250 \frac{d^2 y}{dt^2} + 4108 \frac{dy}{dt} + 1.1202 \times 10^4 y(t) = 5 \times 10^4 f(t)$$

First, find the characteristics equation and the poles:

$$2s^4 Y(s) + 52s^3 Y(s) + 6250s^2 Y(s) + 4108s Y(s) + 1.1202 \times 10^4 Y(s) = 5 \times 10^4 F(s)$$

Characteristic equation is:

$$2s^4 + 52s^3 + 6250s^2 + 4108s + 1.1202 \times 10^4 = 0$$

Poles are:

$$s = -12.68 \pm j5.28 \quad \text{and} \quad s = -0.323 \pm j1.3$$

The dominant pole pair is: $s = -0.323 \pm j1.3$

Since, dominant poles are complex pair, the unit-step response will be under-damped with oscillatory transient response. The overall step response can be approximated based on the dominant poles.

Thus, we can find the damping ratio and the undamped natural frequency from the dominant poles:

$$s = -0.323 \pm j1.3 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The damping ratio is obtained as:

$$\zeta = \cos \theta = \cos \left[\tan^{-1} \left(\frac{\text{Im}[s]}{\text{Re}[s]} \right) \right] = \cos \left[\tan^{-1} \left(\frac{1.3}{0.323} \right) \right] = 0.241$$

The undamped natural frequency is obtained as:

$$\omega_n = \sqrt{\text{Re}[s]^2 + \text{Im}[s]^2} = \sqrt{0.323^2 + 1.3^2} = 1.34 \text{ rad/s}$$

The DC-gain (steady-state response value) of the system is:

$$K = \lim_{s \rightarrow 0} sG(s) = \frac{5 \times 10^4}{1.1202 \times 10^4} = 4.46$$

The damped natural frequency is:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 1.34 \sqrt{1 - 0.241^2} = 1.3 \text{ rad/s}$$

The estimated step response based on the dominant poles will be:

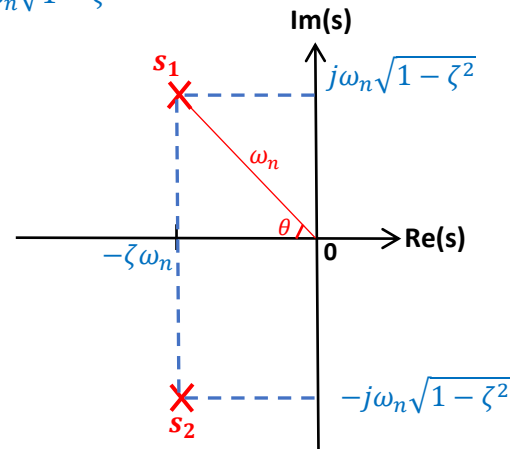
$$y(t) = K - \frac{K e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} (\sin(\omega_d t + \cos^{-1}\zeta)) \quad \rightarrow \quad y(t) = 4.46 - \frac{4.46 e^{-0.323t}}{\sqrt{1 - 0.241^2}} (\sin(1.3t + 76.05^\circ))$$

$$y(t) = 4.46 - 4.6 e^{-0.323t} (\sin(1.3t + 76.05^\circ))$$

The settling time from the dominant poles is:

$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{0.323} = 12.38 \text{ sec}$$

which represent the required time for response to reach the steady-state.



- 6) Use a software package such as MATLAB to plot the step response of the following model for three cases: $a = 0.2$, $a = 1$ and $a = 10$. The step input has a magnitude of 2500.

$$\frac{d^4 y}{dt^4} + 24 \frac{d^3 y}{dt^3} + 225 \frac{d^2 y}{dt^2} + 900 \frac{dy}{dt} + 2500 y(t) = f(t) + a \frac{df}{dt}$$

Compare the response to that predicted by the maximum overshoot, peak time, rise time, and 2% settling time calculated from the dominant poles.

First, find the transfer function model of the system,

$$s^4 Y(s) + 24s^3 Y(s) + 225s^2 Y(s) + 900s Y(s) + 2500 = F(s) + a s F(s)$$

$$G(s) = \frac{Y(s)}{F(s)} = \frac{as + 1}{s^4 + 24s^3 + 225s^2 + 900s + 2500}$$

The MATLAB code to plot the step response and find the time response specifications is:

```
den = [1 24 225 900 2500];
```

```
num1 = 2500*[0.2 1];
sys1 = tf(num1,den);
figure; stepplot(sys1)
stepinfo(sys1)
```

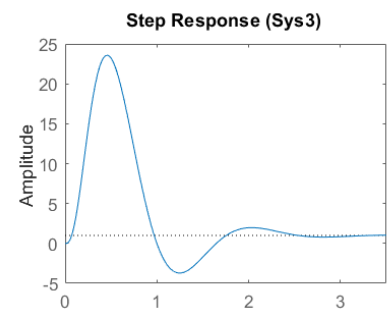
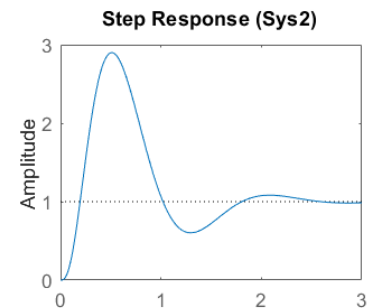
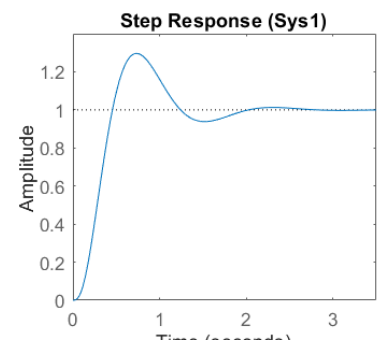
```
num2 = 2500*[1 1];
sys2 = tf(num2,den);
figure; stepplot(sys2)
stepinfo(sys2)
```

```
num3 = 2500*[10 1];
sys3 = tf(num3,den);
figure; stepplot(sys3)
stepinfo(sys3)
```

The step response specifications are obtained as:

a	Overshoot	Peak time	Rise time	Settling time
0.2	30%	0.7368	0.2848	1.9
1	190%	0.5066	0.1130	2.9
10	2256%	0.4605	0.0385	2.3

We can estimate the system as a second-order system based on the dominant poles and compare the step response specifications with the obtained values for MATLAB simulation.



First, find the characteristics equation and the poles:

$$s^4 Y(s) + 24s^3 Y(s) + 225s^2 Y(s) + 900s Y(s) + 2500 = F(s) + asF(s)$$

Characteristic equation is:

$$s^4 + 24s^3 + 225s^2 + 900s + 2500 = 0$$

Poles are:

$$s = -10 \pm j5 \quad \text{and} \quad s = -2 \pm j4$$

The dominant pole pair is: $s = -2 \pm j4$

We can find the damping ratio and the undamped natural frequency from the dominant poles:

$$s = -2 \pm j4 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The damping ratio is obtained as:

$$\zeta = \cos \theta = \cos \left[\tan^{-1} \left(\frac{\text{Im}[s]}{\text{Re}[s]} \right) \right] = \cos \left[\tan^{-1} \left(\frac{4}{2} \right) \right] = 0.447$$

The undamped natural frequency is obtained as:

$$\omega_n = \sqrt{\text{Re}[s]^2 + \text{Im}[s]^2} = \sqrt{4 + 16} = \sqrt{25} = 5 \text{ rad/s}$$

The settling time from the dominant poles is about:

$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{2} = 2 \text{ sec}$$

The maximum overshoot is estimated as:

$$O.S. \% = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = 0.21 \times 100\% = 21\%$$

The peak-time and rise-time are estimated as:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{5\sqrt{1-(0.447)^2}} = 0.702 \text{ sec}$$

$$t_r = \frac{0.8 + 2.5\zeta}{\omega_n} = \frac{0.8 + 2.5(0.447)}{5} = 0.384 \text{ sec}$$

Compare these values with the MATLAB simulation results, we can conclude that the dominant pole approximation is valid for small values of a , which means that the zero at $-1/a$ is far enough from the dominant poles.

